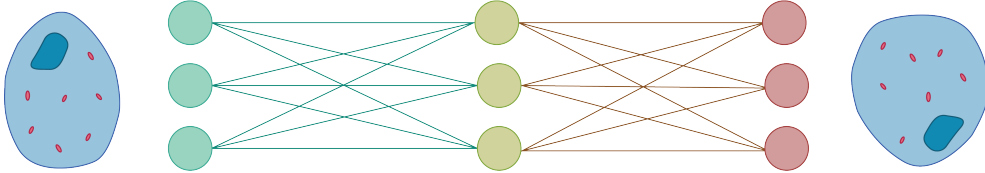


Graph Representation Learning in Cell Biology

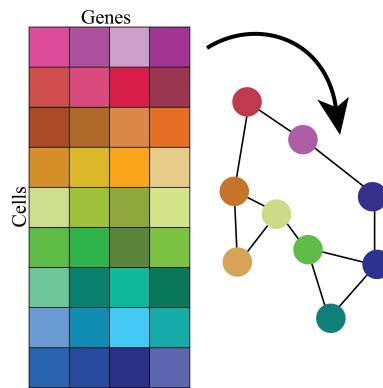
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1 Single-Cell RNA Sequencing Data

Though mathematics and biology have long been disparate fields, computer scientists, mathematicians and biologists are beginning to collaborate. They are taking advantage of the recent technological advancements which have allowed biologists to collect cell data on a single-cell scale, as well as advances in machine learning and mathematics which allow computers to process these massive data sets. To collect single-cell data, biologists use RNA sequencing, which enables scientists to study the set of all messenger RNA (mRNA) molecules in a cell also known as the transcriptome. This technology can be used to compare individual cell transcriptomes, so scientists can discern the heterogeneity between cells, which is immensely useful in cancer research. Additionally, single-cell RNA sequencing data provides information on gene expression at the single-cell level.

From this data, a matrix can be constructed in which rows refer to individual cells, while columns refer to genes. A k -nearest neighbors graph can be constructed by giving a vertex on the graph the labels of its k nearest neighbors. In this case, the nearest neighbors are found by a genetic similarity metric, resulting in the connection of cells that are genetically similar.



2 Graph Representation Learning

Neural networks are networks of nodes or artificial neurons that each have a threshold. When the input these nodes receive exceeds their threshold value, they return an output. Each node also has a weight which determines how strong its output is. After the network of nodes receives an input, it sums its outputs and passes it to the next layer or network of nodes. These networks train on labeled sets, comparing their output to the label of the input to determine the optimal weights for their nodes. Using the vast data generated by single-cell RNA sequencing as a training set for neural networks allows these networks to make predictions on biological factors that can help predict cancer types.

To achieve this functionality, nodes on the graph are mapped to vector space, also referred to as latent space. Graph representation learning can occur in either an unsupervised, supervised, or semi-supervised setting. In an unsupervised setting, the most relevant property of the graph for the question at hand must be identified. In this way, unsupervised learning analyzes and clusters a data set and discovers patterns without manual help. Supervised learning, on the other hand, uses labeled data sets and can sort data into categories or analyze the relationships between dependent and independent variables. Semi-supervised learning combines these two, using a small set of labeled training data and a larger set of unlabeled data. In the context of RNA-sequencing data, unsupervised learning can create cellular similarity graphs without human intervention, while semi-supervised learning can classify cancers by using a labeled set of cancerous cells.

Graph neural networks accept a graph as an input and pass a graph as the output, updating the information in the nodes and edges without changing the actual connectivity of the input graph. Because of how broad graphs are, it is difficult to represent them in a format that computers can accept. To tackle this problem, the success of convolutional neural networks can be generalized from grid-like structures to arbitrary graphs. This generalization can be achieved through polynomial filters on graphs.

Before we proceed, let's define some terms:

- An **adjacency matrix** is a square matrix that can be used to represent a graph. The elements of the matrix indicate the adjacency of the vertices on the graph. For any two vertices on the graph, there exists an entry in the matrix that indicates if there exists an edge between the vertices (1 for an edge, 0 for no edge).
- For some graph G with adjacent matrix A , the **diagonal degree matrix** D can be constructed with:

$$D_{vu} = \sum_u A_{vu}$$

where A_{vu} denotes the entry in row v and column u of adjacency matrix A .

- The **Graph Laplacian** L is a square matrix defined as $L = D - A$. From the graph Laplacian, polynomials of the form:

$$w_0 I_n + w_1 L + w_2 L^2 + \dots + w_d L^d$$

can be constructed. These polynomials act as filters, with the coefficients behaving as the weights of the filters. To perform a convolution on a graph, nodes can be stacked to create a vector x . A convolution x' upon the graph with vector x can be expressed as

$$x' = p_w(L)x$$

This means that convolutions at node v occur only with nodes u that are no more than d steps away. The attributes of a node are combined with those of its immediate neighbors through this process, based on the coefficients and degree of the polynomial. To construct a graph neural network from this idea of convolving by a polynomial filter, the coefficients or weights of the filter are set as learnable parameters for the neural network.

3 Applications

Graph convolutional networks are powerful tools capable of deciphering large datasets. An example of this capability is a study which analyzed 913 cells from the cortex of a mouse, including thousands of interactions between ligands and receptors. Ligands are molecules or atoms that bind to a protein, transmitting a signal. Graph convolutional networks were shown to be 20% more effective than previous methods in detecting receptor-ligand interactions from spatial data.

Additionally, graph convolutional networks can be used to predict the directionality of other interactions. Using ligand-receptor interactions whose directionality is known (the ligand interacts with the receptor, not the other way around), graph convolutional networks were able to discern the directionality of other pairs.

Finally, graph convolutional networks have also been used to predict the function of genes. Proteins have a similar function as proteins they interact with, leading to the “guilt by association” method of predicting protein function. Graph neural networks serve as an alternative, using spatial data to predict protein associations. Since Genes are composed of proteins, both of these methods can be applied to genes as well. Convolutional graph networks outperform “guilt by association” predictions by a large margin. Being able to accurately predict the function of genes or of biological interactions is important because they allow scientists and doctors to better treat patients and develop medications.

4 References

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